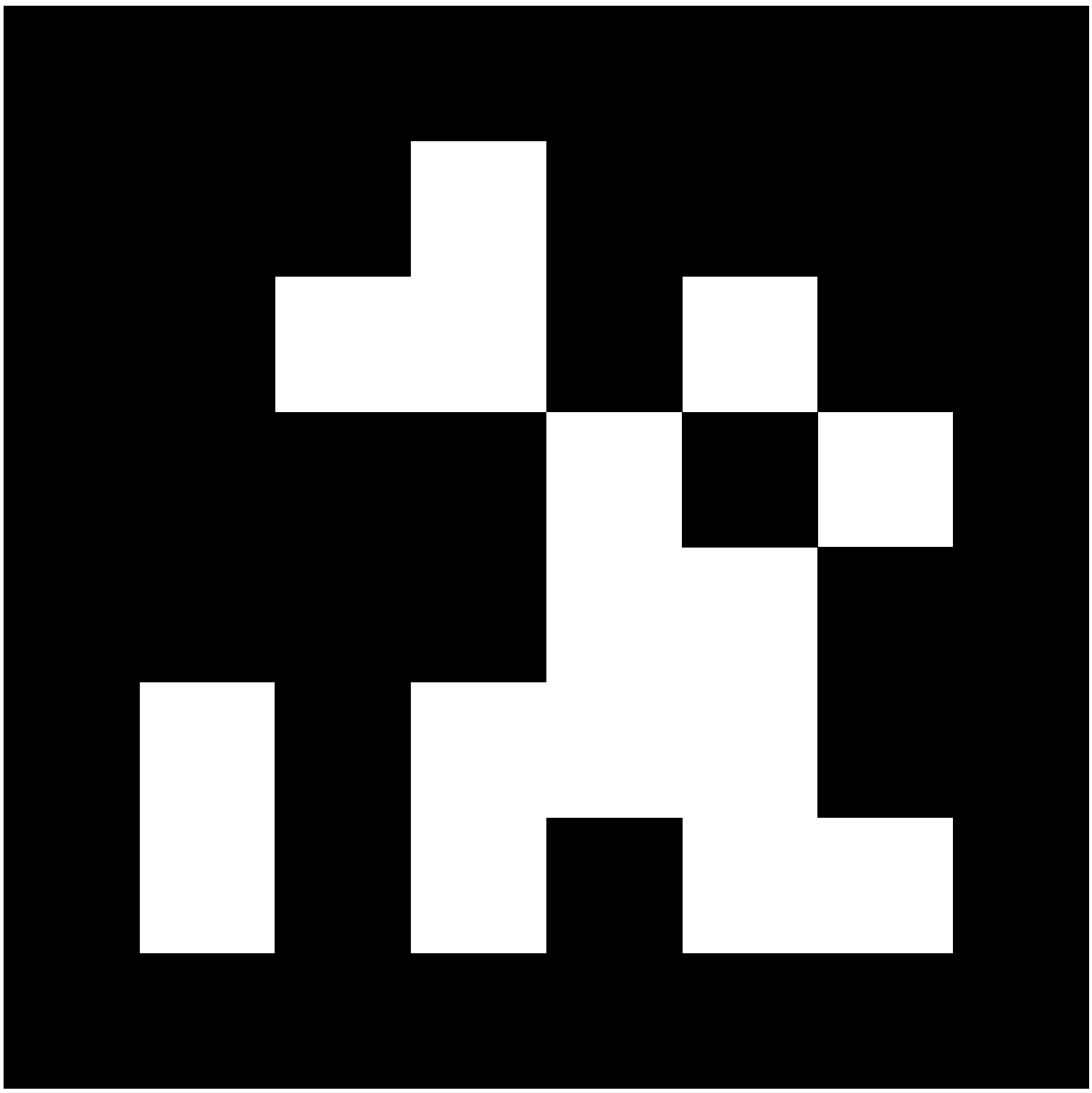
start

AprilTag 36h11 ID 0



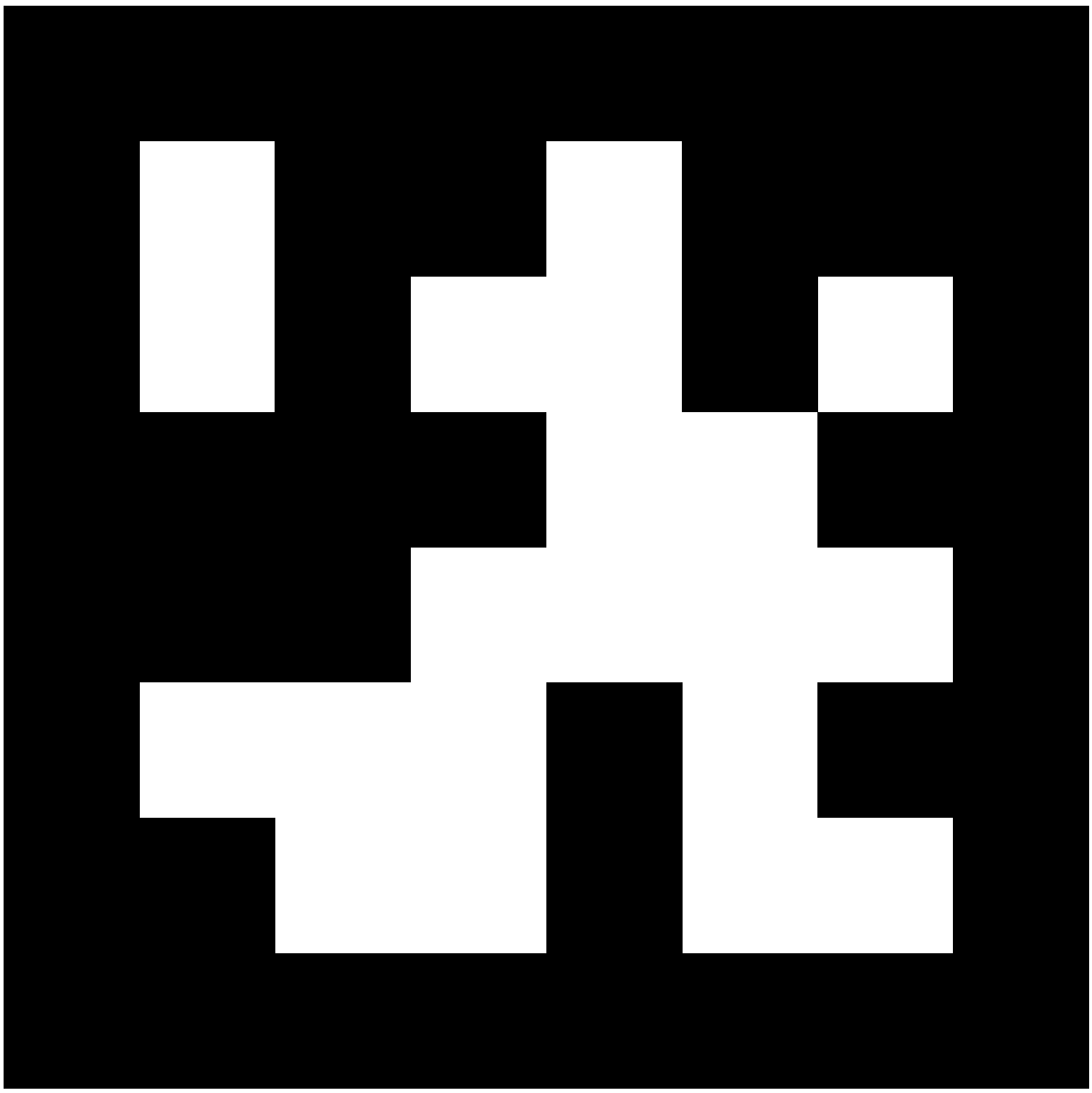
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Start/finish line, on the right-hand (north) lane wall; the truck starts beside it.

|<-- this bar is 100 mm -->|

ramp_up

AprilTag 36h11 ID 1



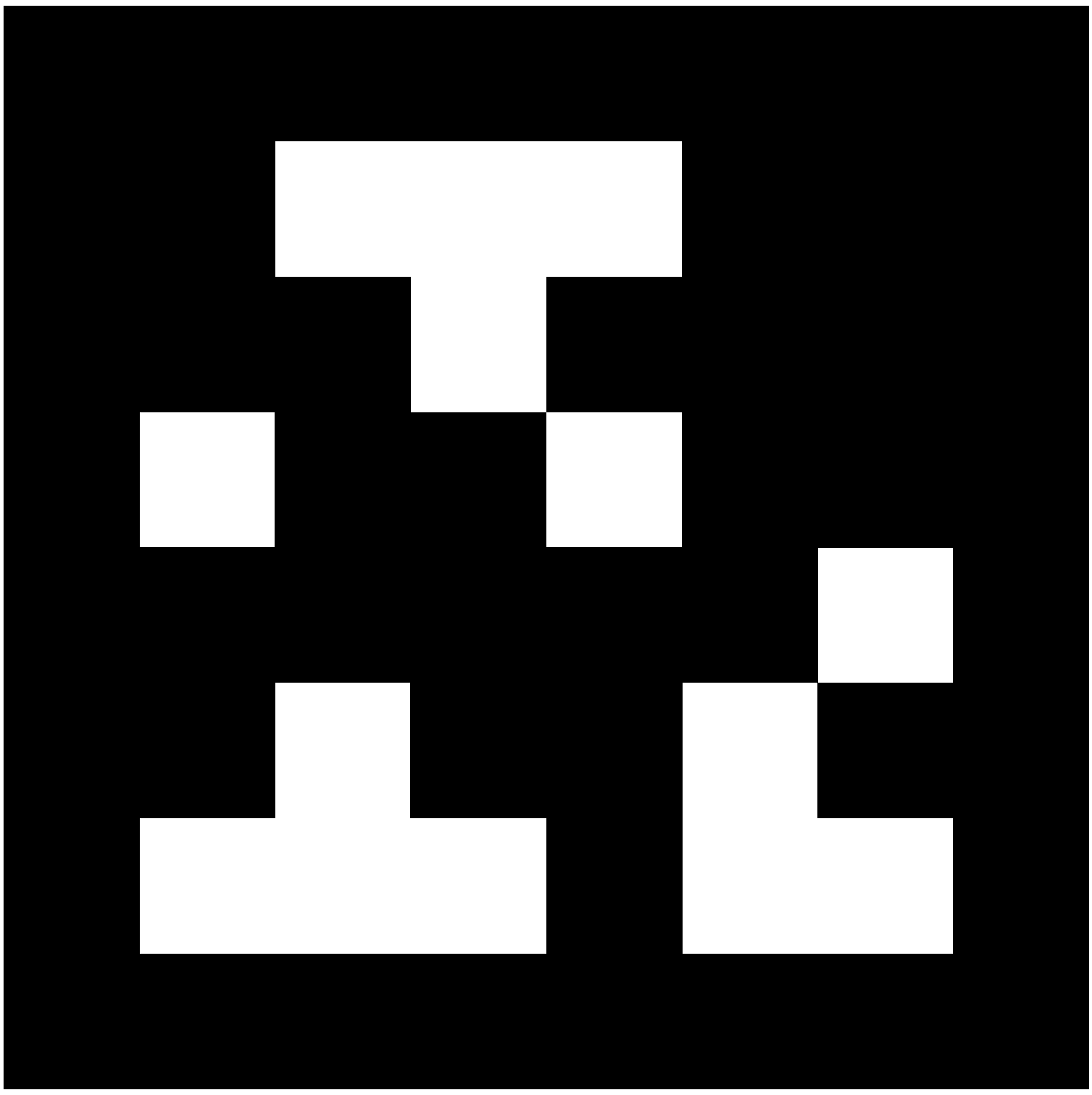
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Foot of the ramp (3.33 m past the start line), right-hand wall line.

|<-- this bar is 100 mm -->|

tunnel

AprilTag 36h11 ID 2



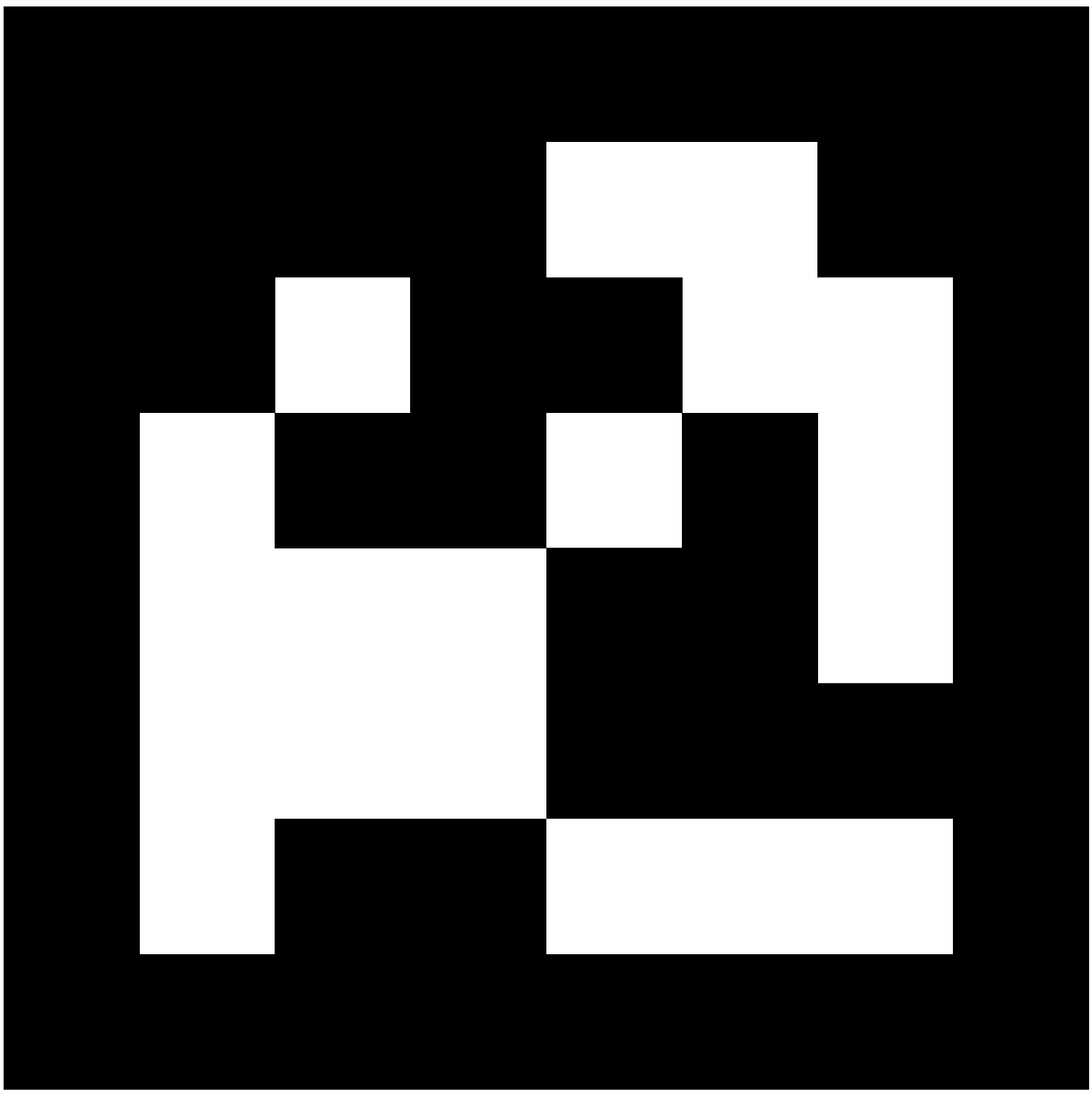
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Helix exit, where the 32" funnel into the tunnel starts; right-hand (east) wall.

|<-- this bar is 100 mm -->|

narrow

AprilTag 36h11 ID 3



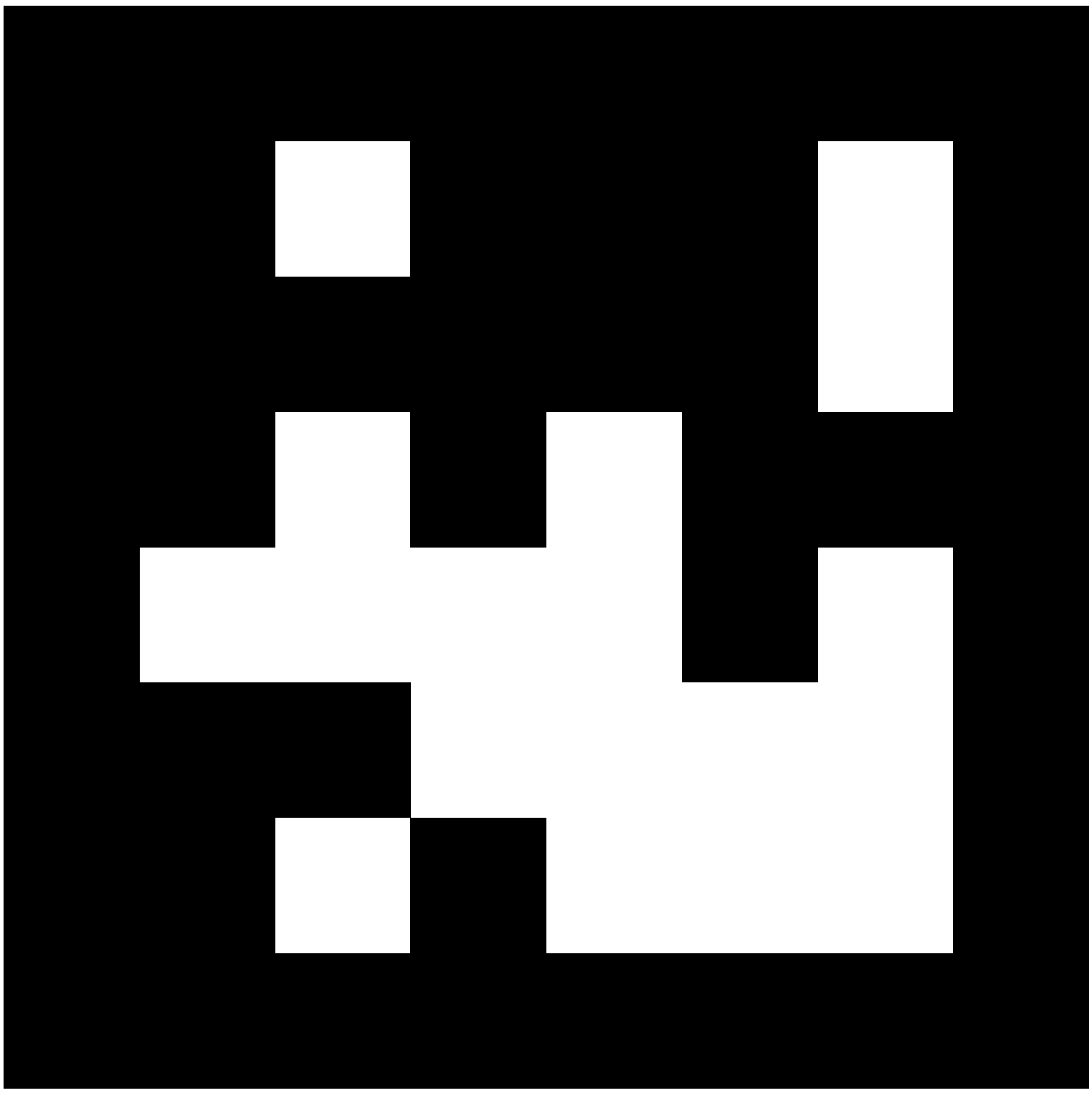
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

End of the tunnel straight, where the lane narrows 32" -> 20"; right-hand (east) wall.

|<-- this bar is 100 mm -->|

gravel

AprilTag 36h11 ID 4



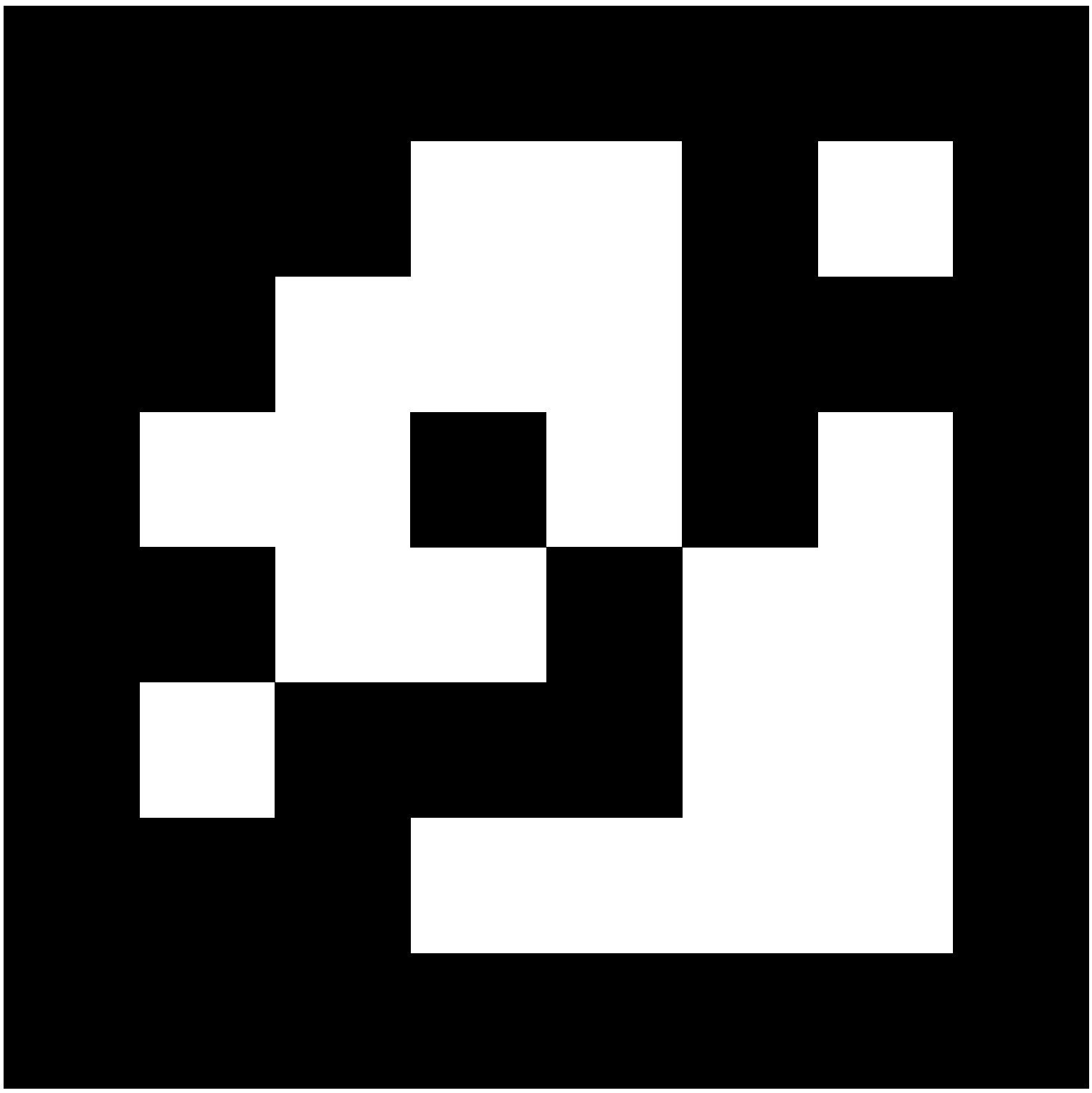
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Foot of the gravel ramp-up (start of the 32" gravel entry); right-hand (south) wall.

|<-- this bar is 100 mm -->|

bank_approach

AprilTag 36h11 ID 5



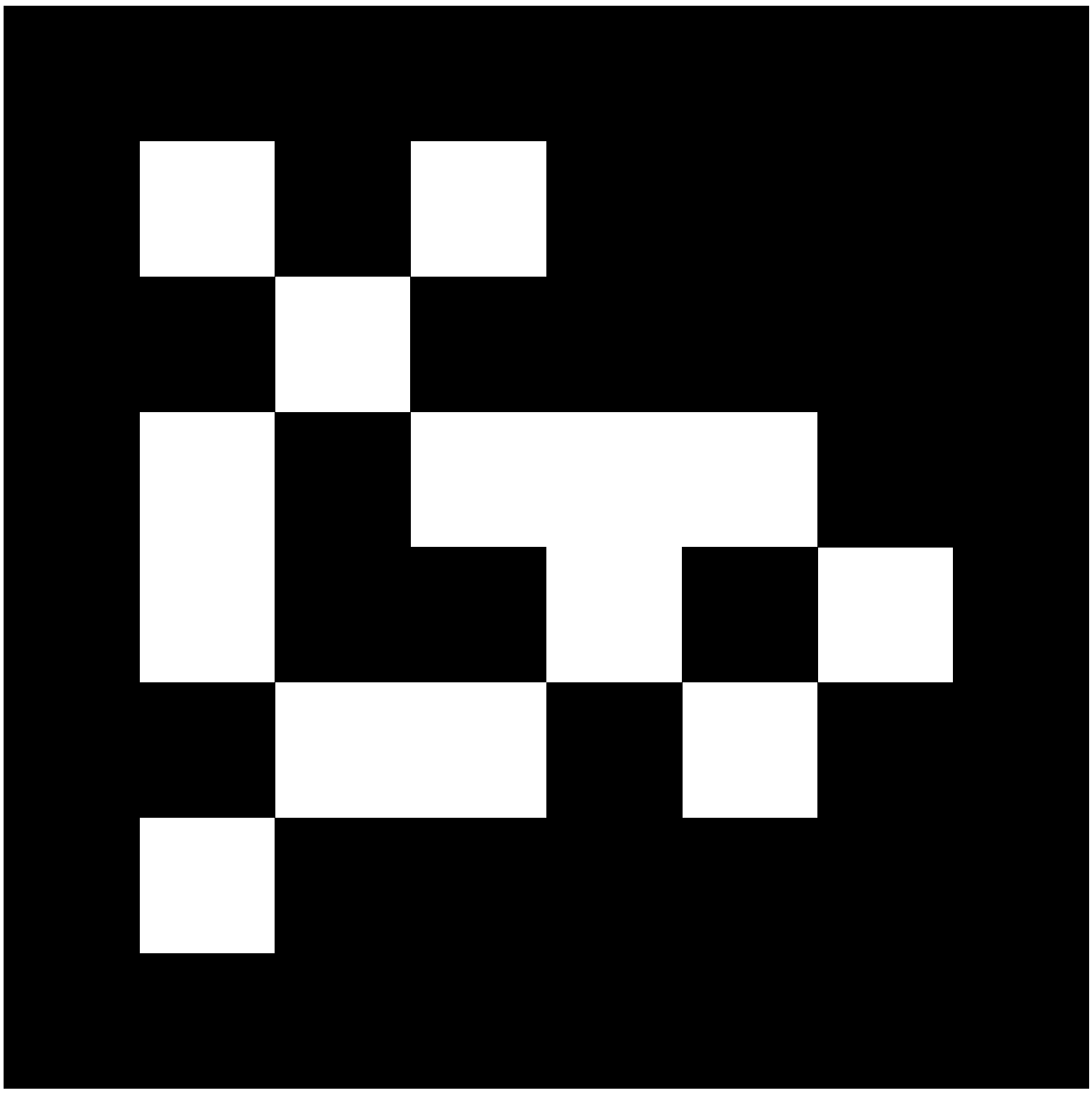
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Zone line at the start of the bank entry lane; right-hand (south) side, 5 cm outside the lane wall.

|<-- this bar is 100 mm -->|

pothole

AprilTag 36h11 ID 6



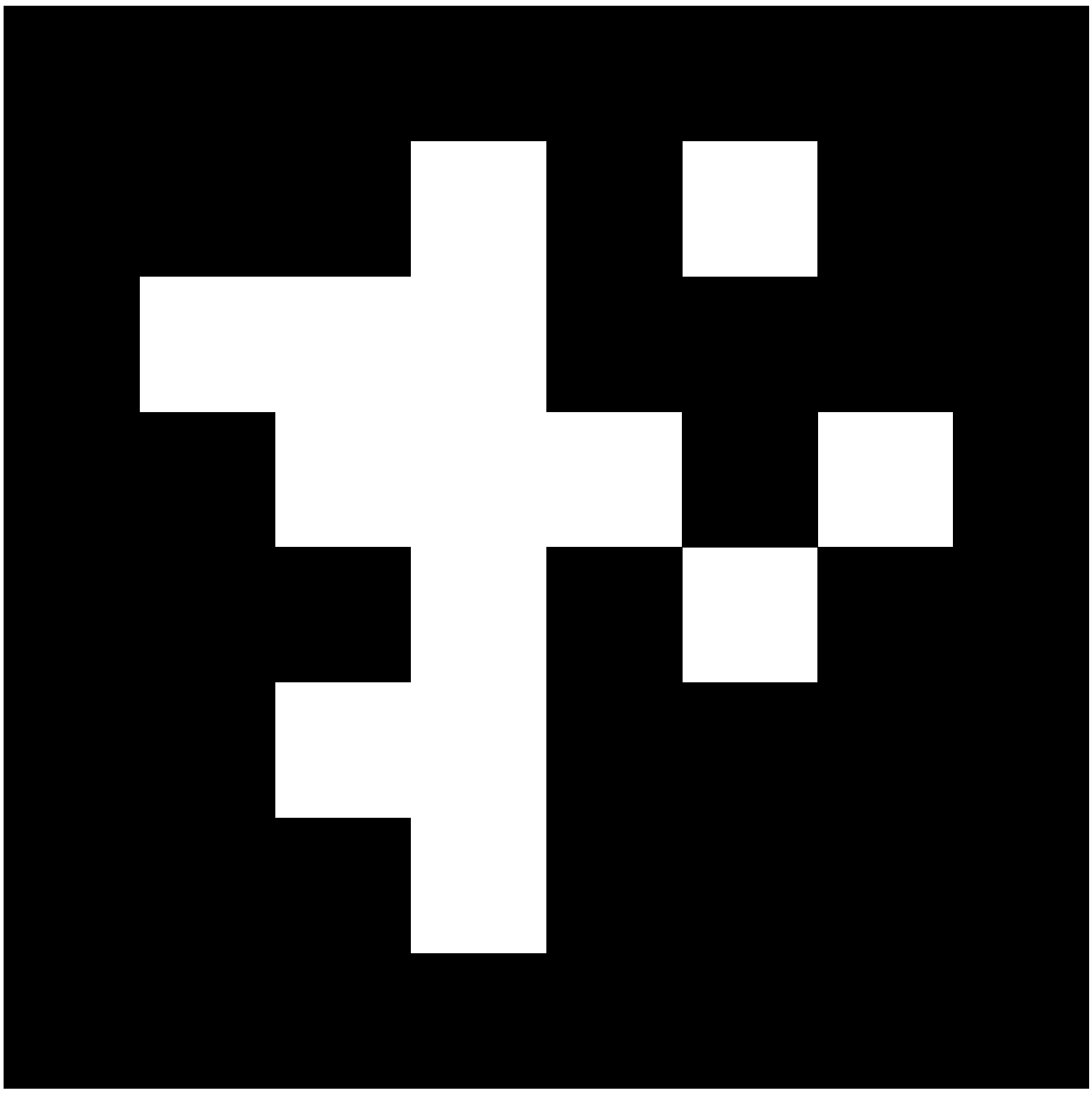
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Zone line at the start of the return lane (west of the bank); right-hand (north) wall.

|<-- this bar is 100 mm -->|

wide

AprilTag 36h11 ID 7



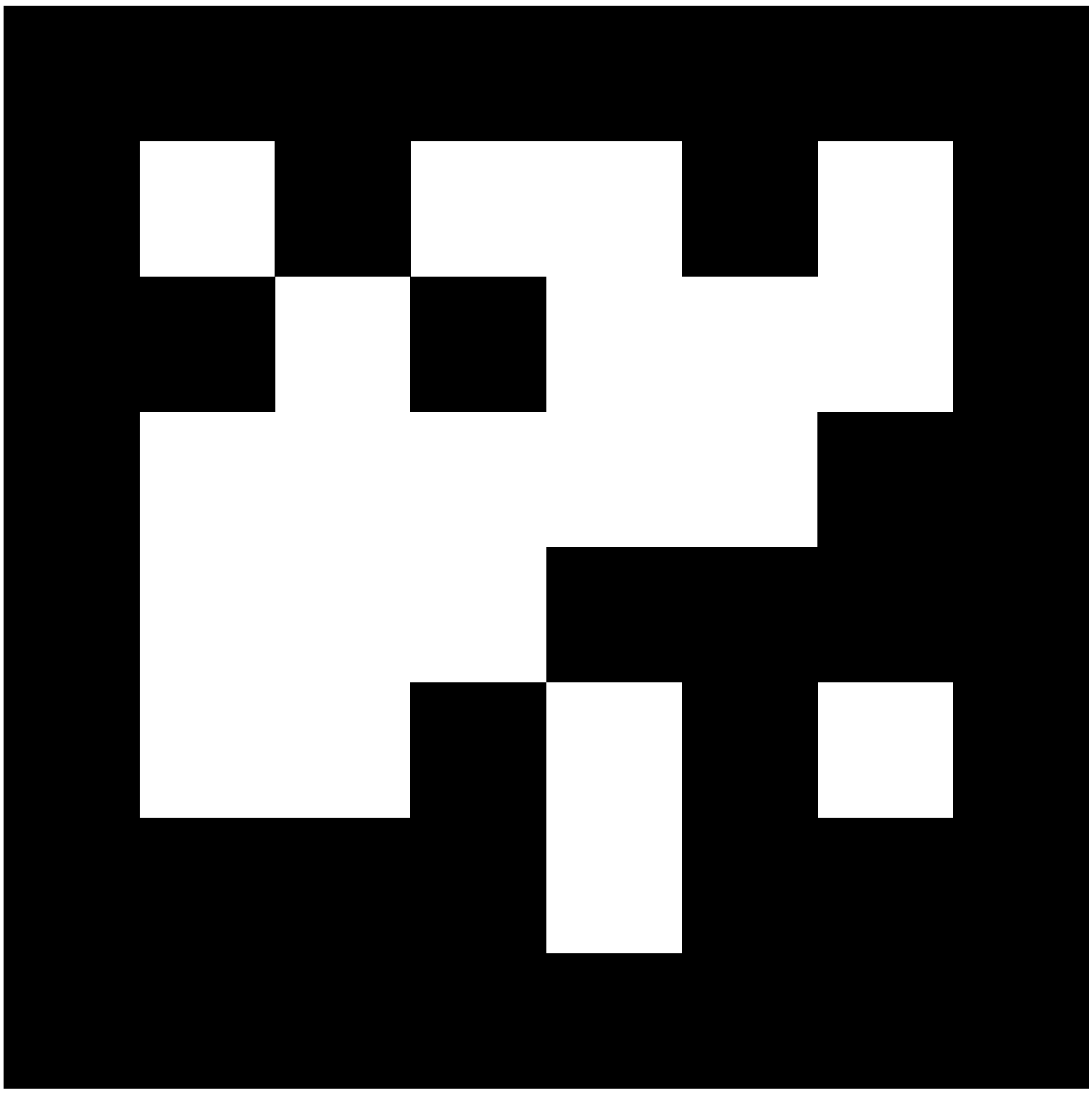
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

See placement.csv / placement_map.png

|<-- this bar is 100 mm -->|

buckets

AprilTag 36h11 ID 8



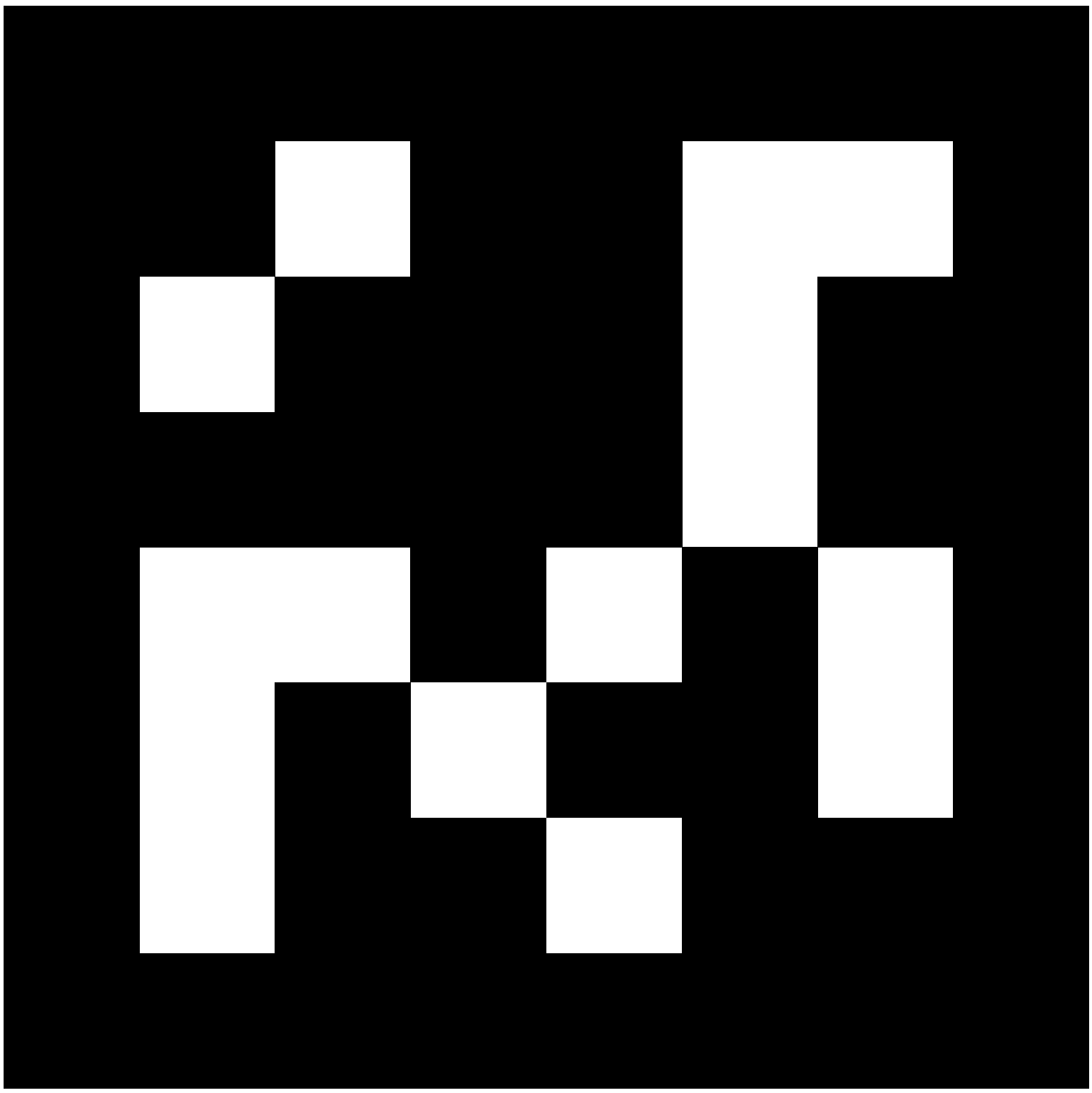
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

On the wide-area wall that holds the bucket-field entrance, 0.5 m to the right (south) of the entrance gap centre.

|<-- this bar is 100 mm -->|

hoops

AprilTag 36h11 ID 9



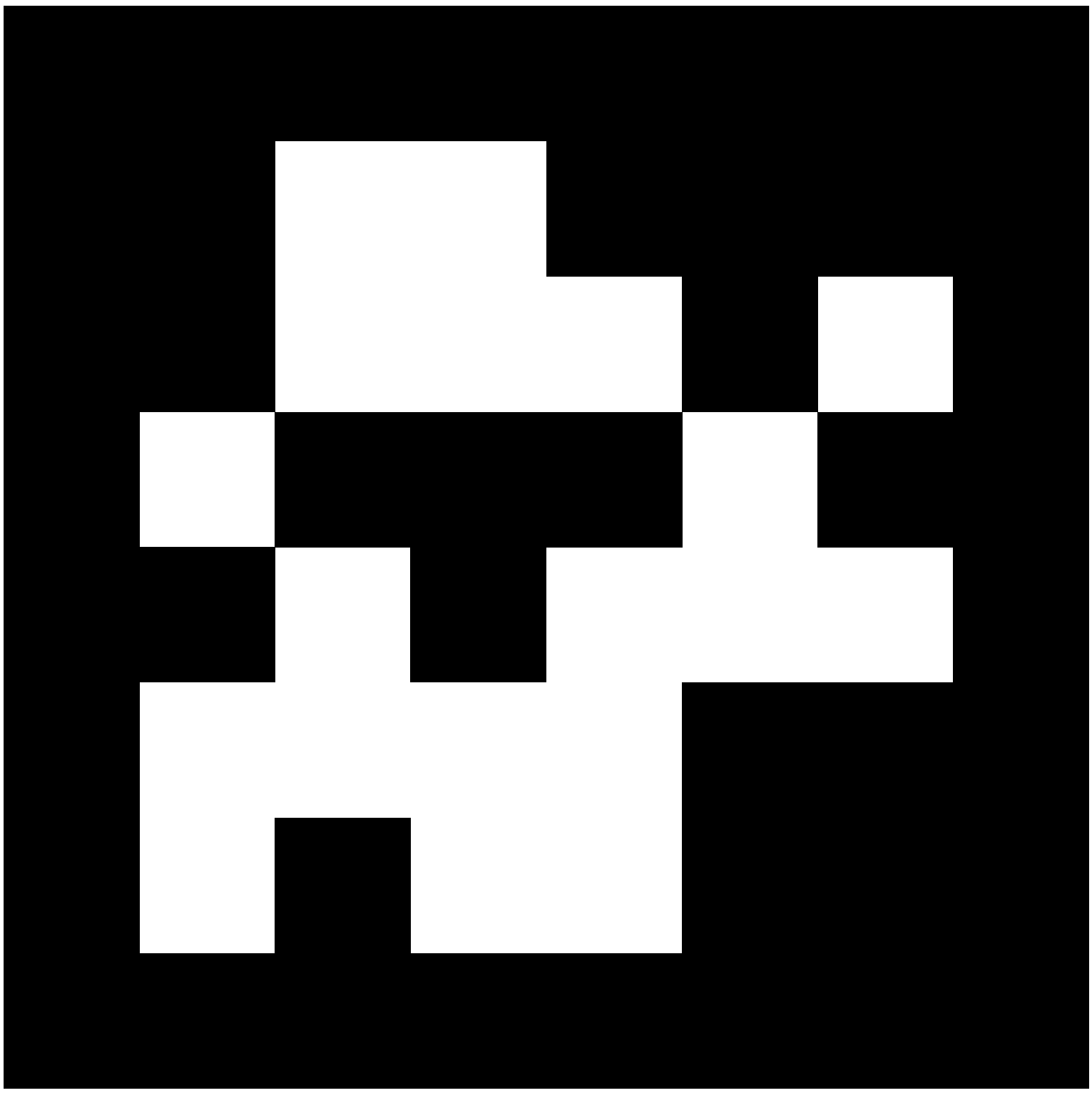
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

On the bucket-field far wall, 0.5 m to the right (south) of the exit gap centre (start of the 36" connector).

|<-- this bar is 100 mm -->|

car_wash

AprilTag 36h11 ID 10



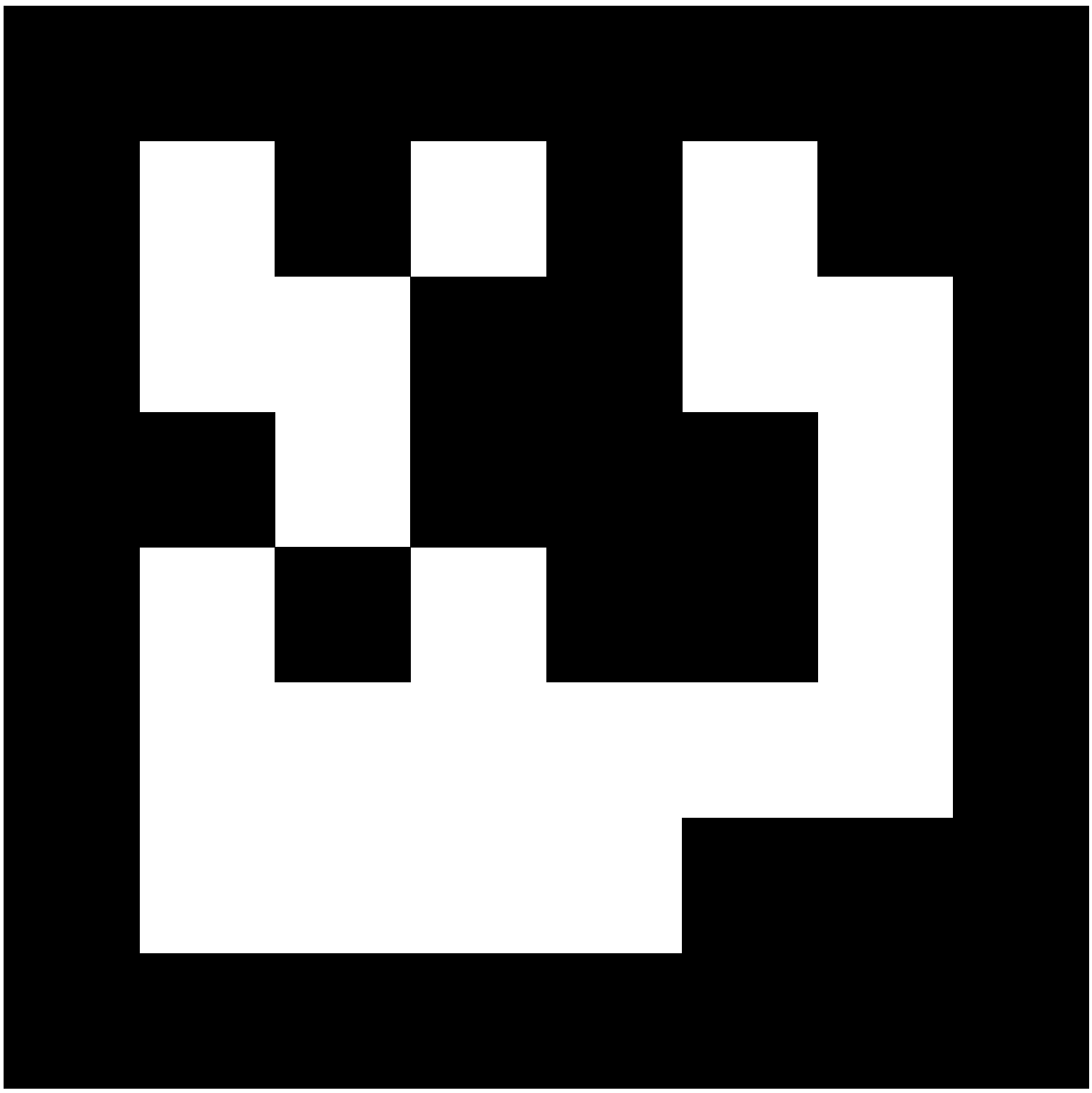
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

See placement.csv / placement_map.png

|<-- this bar is 100 mm -->|

finish

AprilTag 36h11 ID 11



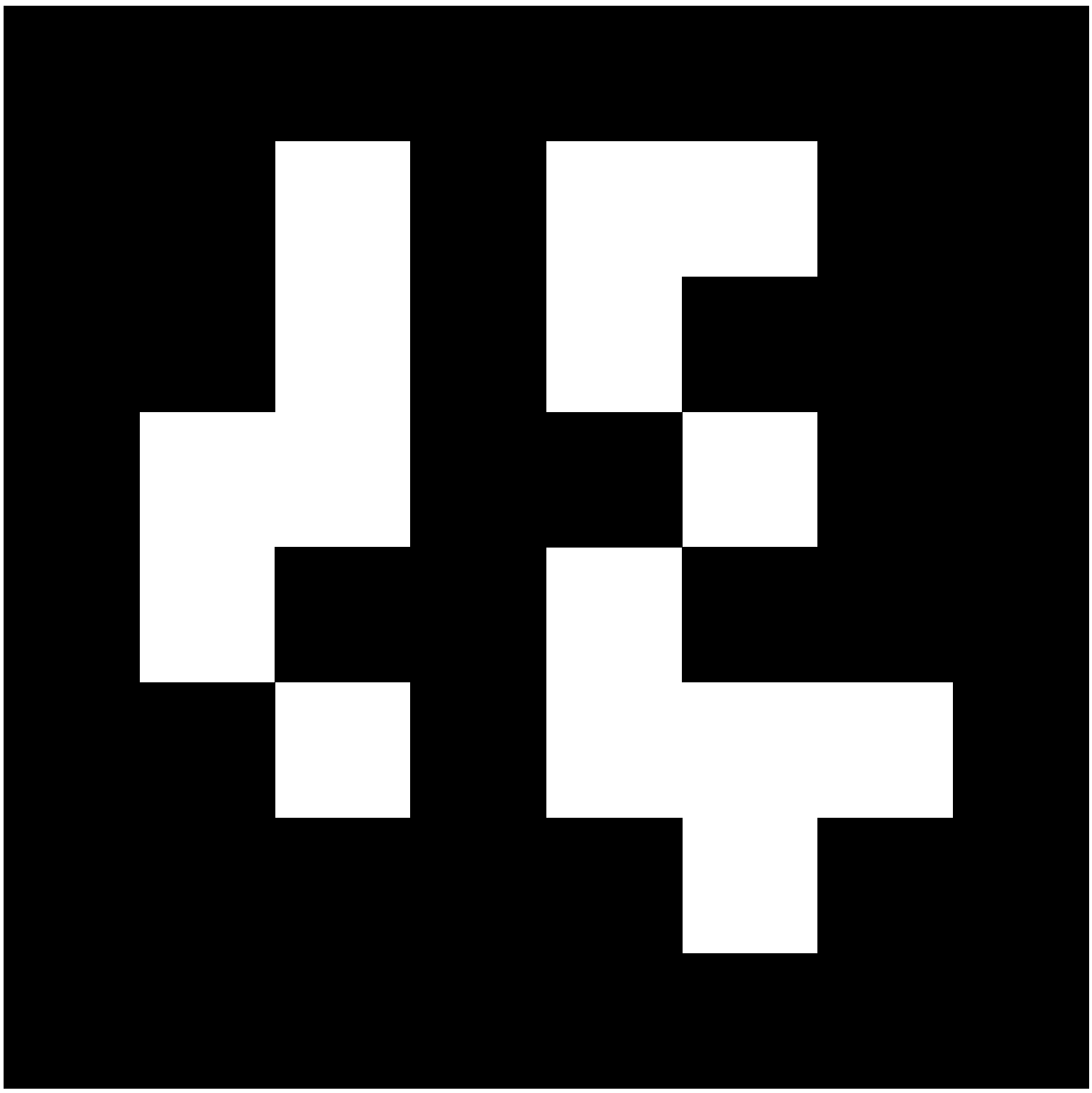
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

0.85 m before the finish line inside the 32" funnel, right-hand (north) side.

|<-- this bar is 100 mm -->|

speed_1

AprilTag 36h11 ID 12



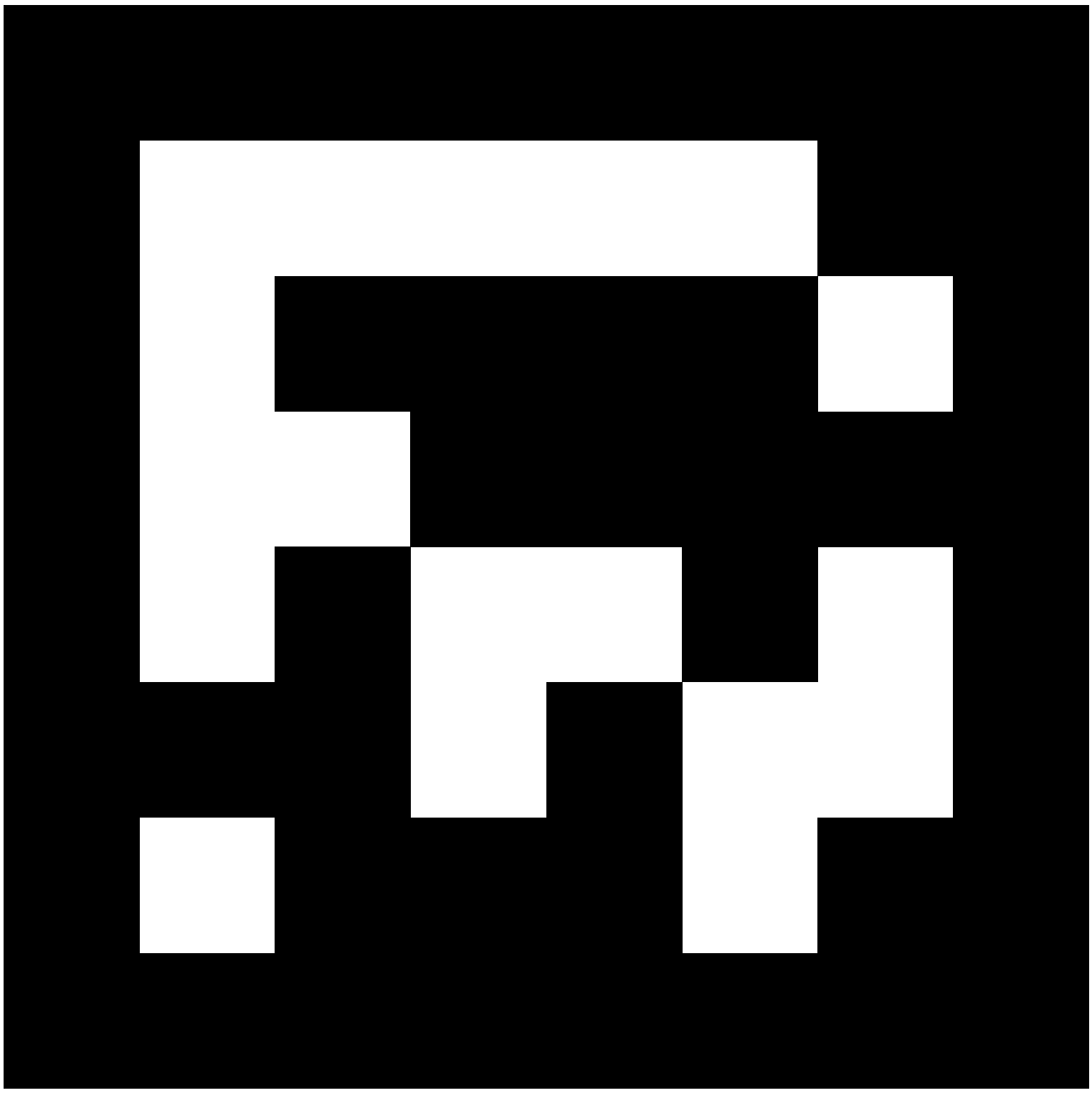
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Speed course (reserved: not in marker_detector.tag_ids, so the obstacle mission ignores it).

|<-- this bar is 100 mm -->|

speed_2

AprilTag 36h11 ID 13



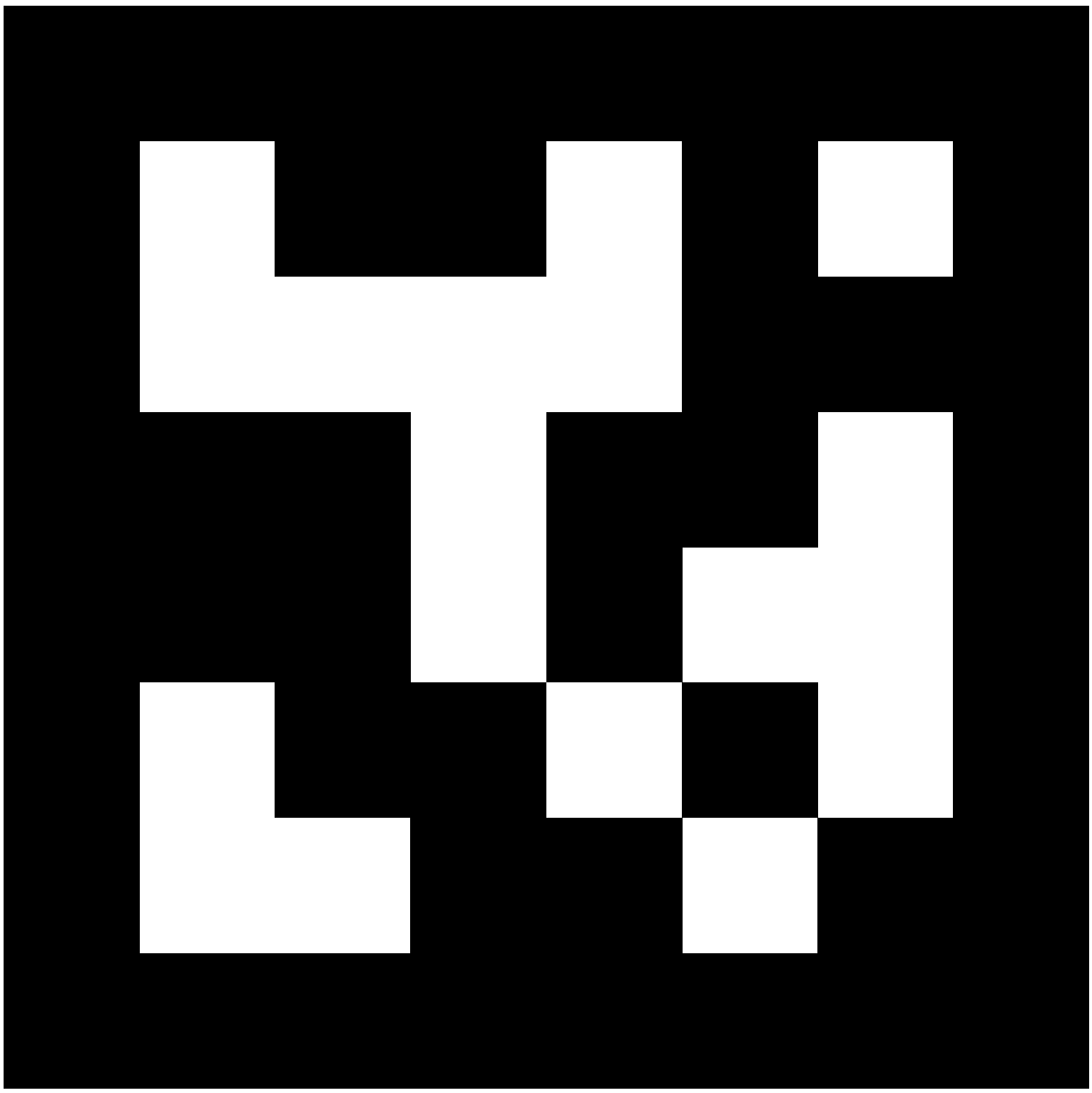
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Speed course (reserved: not in marker_detector.tag_ids, so the obstacle mission ignores it).

|<-- this bar is 100 mm -->|

speed_3

AprilTag 36h11 ID 14



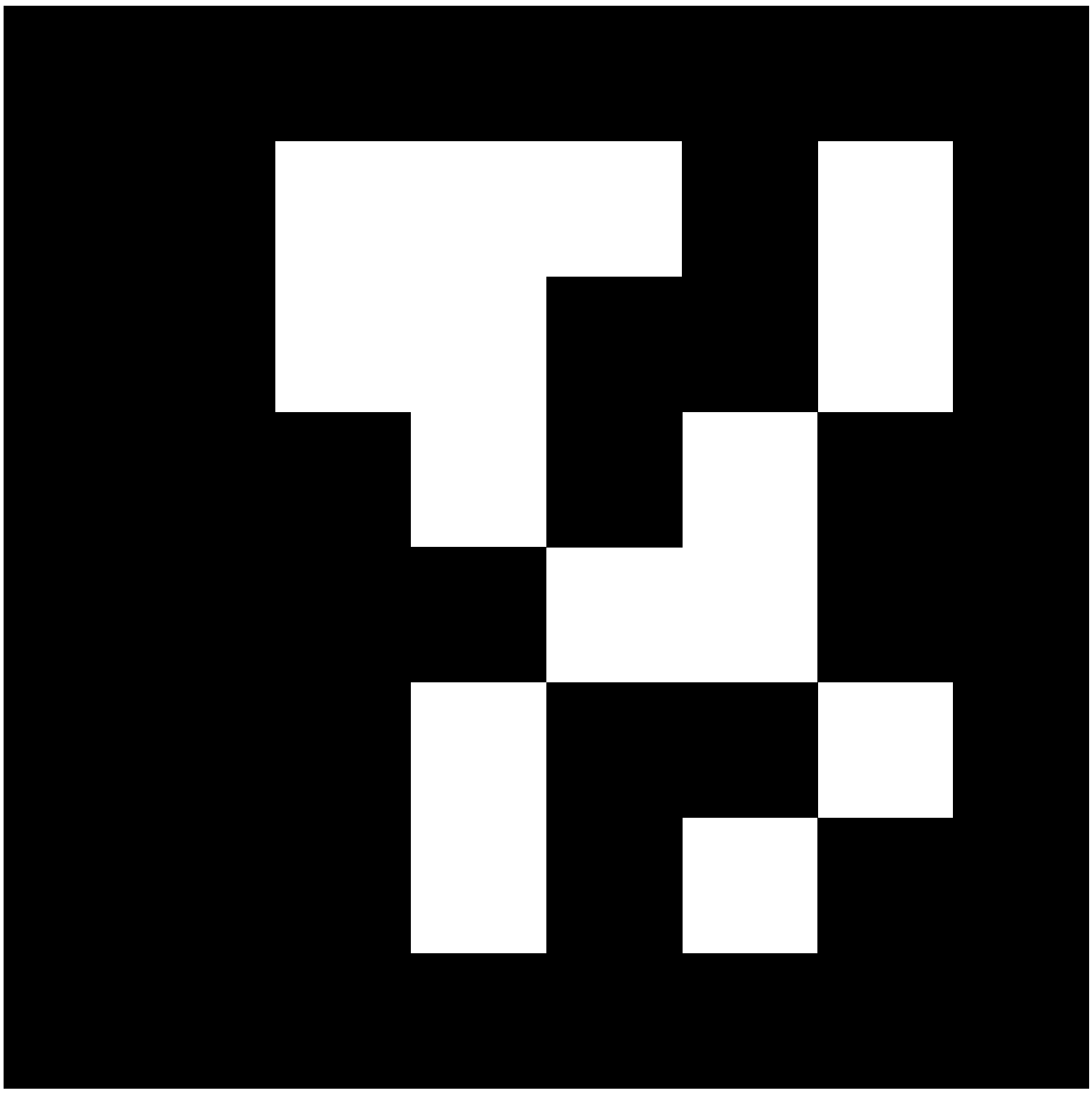
Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Speed course (reserved: not in marker_detector.tag_ids, so the obstacle mission ignores it).

|<-- this bar is 100 mm -->|

speed_4

AprilTag 36h11 ID 15



Black square must measure 200 mm. Print A3 at 100% / actual size on matte paper; keep at least 25 mm of white around the black square.

Speed course (reserved: not in marker_detector.tag_ids, so the obstacle mission ignores it).

|<-- this bar is 100 mm -->|